

Linux Lab 7

By Jai Pandey (260840140011)

Question 1: Setting up a Web Server

1. Installing httpd

```
[root@localhost named]# dnf install httpd
Last metadata expiration check: 0:04:24 ago on Friday 04 September 2026 03:19:04 PM.
Package httpd-2.4.62-15.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost named]#
```

2. Adding the base domain and www type A records

```
csf@localhost: / — vim /var/named/for.demo.lab

$TTL 1D
@      IN SOA  csf.demo.lab. root@gmail.com. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H      ; minimum
@      NS   csf.demo.lab.
csf     A    192.168.200.100
www     A    192.168.200.100
demo.lab A    192.168.200.100
portal  CNAME 192.168.200.100
~
~
```

3. Restarted the named service

```
[root@localhost /]# systemctl restart named
[root@localhost /]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-09-04 15:27:08 IST; 7s ago
     Process: 35090 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ];
     Process: 35092 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exi
   Main PID: 35093 (named)
      Tasks: 14 (limit: 48546)
     Memory: 38.9M (peak: 39.7M)
        CPU: 309ms
    CGroup: /system.slice/named.service
            └─35093 /usr/sbin/named -u named -c /etc/named.conf

Sep 04 15:27:08 csf named[35093]: network unreachable resolving './DNSKEY/IN': 2001:dc3:
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './NS/IN': 2001:dc3::35#
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './DNSKEY/IN': 2001:7fd:
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './NS/IN': 2001:7fd::1#5
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './DNSKEY/IN': 2001:503:
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './NS/IN': 2001:503:ba3e
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './DNSKEY/IN': 2001:500:
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './NS/IN': 2001:500:2::c
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './DNSKEY/IN': 2001:503:
Sep 04 15:27:08 csf named[35093]: network unreachable resolving './NS/IN': 2001:503:c27:
lines 1-22/22 (END)
```

4. Installing PHP

```
[root@localhost html]# dnf install php
Last metadata expiration check: 0:12:14 ago on Friday 04 September 2026 03:19:04 PM.
Package php-8.0.30-8.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

5. Adding index.php files to Apache default document root directory

```
[root@localhost /]# cd /var/www/html/
[root@localhost html]# ll
total 0
[root@localhost html]# vim index.php
```

```
<?php
echo 'Hello World';
echo '
<html>
    <body>
        <h1>"This is a PHP file!"</h1>
    </body>
</html>' ;
?>
~
~
```

6. Restarted the named service

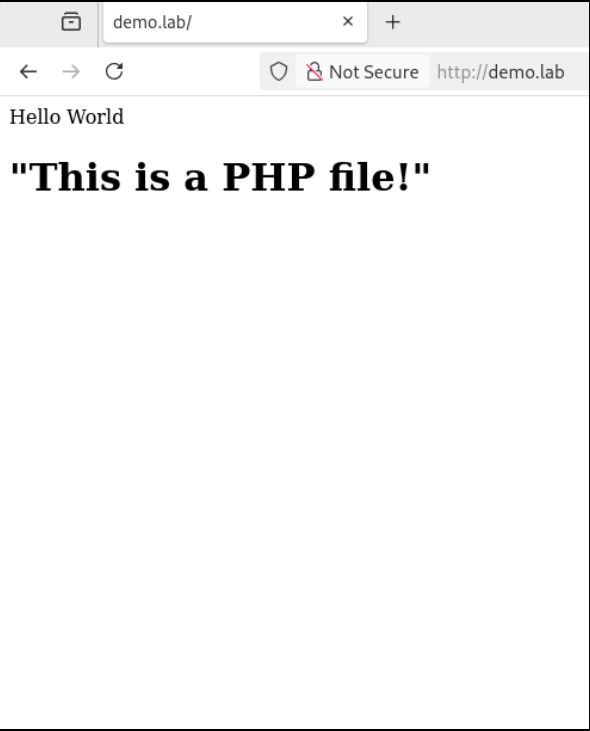
```
[root@localhost html]# systemctl restart named
[root@localhost html]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-09-04 15:32:28 IST; 9s ago
     Process: 35318 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/s
     Process: 35320 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=
   Main PID: 35321 (named)
      Tasks: 14 (limit: 48546)
     Memory: 38.9M (peak: 39.6M)
        CPU: 327ms
     CGroup: /system.slice/named.service
            └─35321 /usr/sbin/named -u named -c /etc/named.conf

Sep 04 15:32:28 csf named[35321]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './DNSKEY/IN': 2001:500:12::d0d#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './DNSKEY/IN': 2001:500:9f::42#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './DNSKEY/IN': 2001:7fe::53#53
Sep 04 15:32:28 csf named[35321]: network unreachable resolving './NS/IN': 2001:7fe::53#53
[root@localhost html]#
```

7. Viewing the PHP page on the browser.

```
[root@localhost html]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Fri 2026-09-04 15:39:34 IST; 1min 1s ago
     Docs: man:httpd.service(8)
   Main PID: 35940 (httpd)
    Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 48546)
     Memory: 15.2M (peak: 15.6M)
        CPU: 424ms
     CGroup: /system.slice/httpd.service
            └─35940 /usr/sbin/httpd -DFOREGROUND
              └─35947 /usr/sbin/httpd -DFOREGROUND
                └─35948 /usr/sbin/httpd -DFOREGROUND
                  └─35949 /usr/sbin/httpd -DFOREGROUND
                    └─35951 /usr/sbin/httpd -DFOREGROUND

Sep 04 15:39:24 csf systemd[1]: Starting The Apache HTTP Server:
Sep 04 15:39:34 csf httpd[35940]: AH00558: httpd:
Sep 04 15:39:34 csf httpd[35940]: Server configured
Sep 04 15:39:34 csf systemd[1]: Started The Apache HTTP Server:
[root@localhost html]#
```

A screenshot of a web browser window. The address bar shows 'demo.lab/' and 'http://demo.lab'. The page content displays 'Hello World' in a small font, followed by a large, bold, black text that reads '"This is a PHP file!"'. The browser's security indicator shows 'Not Secure'.

Question 2

8. Installing MariaDB, MariaDB server & php-mysqli

```
[root@localhost html]# dnf install mariadb mariadb-server
Last metadata expiration check: 0:24:50 ago on Friday 04 September 2026 03:19:04 PM.
Package mariadb-3:10.5.29-5.el9.x86_64 is already installed.
Package mariadb-server-3:10.5.29-5.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost html]#
```

9. Performing MariaDB secure installation

```
[root@localhost html]# mariadb-secure-installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n]
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n]
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n]
... Success!

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
[root@localhost html]#
```

10. Logged in as root

```
[root@localhost html]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

11. Created database name results_db

```
MariaDB [(none)]> CREATE DATABASE results_db;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| results_db |
+-----+
4 rows in set (0.001 sec)

MariaDB [(none)]>
```

12. Created user and granted all permissions to results_db

```
MariaDB [(none)]> CREATE USER 'dbadmin'@'localhost' IDENTIFIED BY 'root';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL ON results_db.* to 'dbadmin'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> █
```

13. Logged in as dbadmin

```
[root@localhost html]# mysql -u dbadmin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

14. Checked the list of databases

```
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| results_db |
+-----+
2 rows in set (0.002 sec)

MariaDB [(none)]> █
```

15. Creating the info table

```
MariaDB [(none)]> use results_db;
Database changed
MariaDB [results_db]> CREATE TABLE info (id INT(2), username VARCHAR(20), email VARCHAR(30));
Query OK, 0 rows affected (0.005 sec)

MariaDB [results_db]> DESC info;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int(2) | YES | | NULL | |
| username | varchar(20) | YES | | NULL | |
| email | varchar(30) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

MariaDB [results_db]>
```

16. Inserted info in the table

```
MariaDB [results_db]> INSERT INTO info (id, username, email)
-> VALUES (1, "csf", "root@mail.com"), (2, "jai", "jai@mail.com"), (3, "navneet", "navneet@mail.com");
Query OK, 3 rows affected (0.002 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [results_db]> SELECT * FROM info;
+-----+-----+-----+
| id | username | email |
+-----+-----+-----+
| 1 | csf | root@mail.com |
| 2 | jai | jai@mail.com |
| 3 | navneet | navneet@mail.com |
+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [results_db]> █
```

17. Updated PHP files

```
[root@localhost html]# vim index.php
[root@localhost html]# cat index.php
<?php
include 'db.php';
$result = $conn->query("select * from info");
while ($row = $result->fetch_assoc()){
    echo $row['id'] . " - " . $row['username'] . " - " . $row['email'] . "<br>";
}
$conn->close();
?>
[root@localhost html]#
```

```
[root@localhost html]# vim db.php
[root@localhost html]# cat db.php
<?php
$conn = new mysqli("localhost", "dbadmin", "root", "results_db");
if ($conn->connect_error){
    die("Connection failed");
}
echo "Connection successful";
?>

[root@localhost html]#
```

18. Restarted httpd

```
[root@localhost html]# systemctl restart httpd
[root@localhost html]# █
```

19. Viewing the table in the browser.

